

Multi-stage economic model predictive control for a gold cyanidation leaching process under uncertainty

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ABSTRACT

In order to dynamically operate the gold cyanidation leaching process (GCLP) under uncertainty, a multi-stage economic model predictive control (EMPC) is proposed for GCLP for the transient and steady-state economic optimization. The proposed multi-stage EMPC is composed of two steps. In the first step, the unmeasurable uncertain parameters are estimated by using Tikhonov regularization based method, so as to avoid amplification and propagation of the noise measurements into the estimation. Based on the estimated results, the scenario tree for multi-stage EMPC is generated from the historical data using a data-driven approach, and the control inputs are obtained from solving the resulting large nonlinear programming problem (NLP) at each sampling point. The resulting uncertainty model and the probability of each scenario are more consistent with the actual industrial GCLP, and the solutions are less conservative. The efficiency of the proposed multi-stage EMPC is verified through a simulated industrial GCLP. Compared with other EMPC methods, including classic EMPC and multi-stage EMPC with box uncertainty region, the proposed method can reduce the economic cost while accounting for the constraints at the same time.

KEYWORDS

EMPC, gold cyanidation leaching process, multi-stage NMPC, scenario tree, uncertainty

1 | INTRODUCTION

The gold cyanidation leaching process (GCLP) has been the most common approach for extracting gold from ores in alkaline cyanide pulp.¹ The GCLP has been successfully applied in many gold smelting plants for more than one century, due to its simple structure, small space occupation, and low operation cost.² In order to speed up the gold recovery, a cascade of continuous stirred tank reactors (CSTRs) is usually used in industrial plants^{3,4} for higher recovery rate, and longer residence time. Since the gold product is of high value, the recovery rate is the most important indicator in gold smelting plants. Generally, the GCLP is operated manually in a heuristic way, and excessive cyanide is added to guarantee the recovery of the gold product.⁵

Due to the increasing global market competition, the incentive to reduce the economic cost of the GCLP is becoming more and more important. In addition, excessive addition of cyanide will have a great

adverse effect on the environment. To this end, advanced process control methods are required to refine the operation of GCLP.⁶ A precise process model is often the key factor for process control and optimization, and the process model of GCLP has been investigated widely in the past few decades.^{7–10} Among these existing models, de Andrade Lima and Hodouin¹⁰ developed a first principle model including the mass conservation equations and kinetic reaction equations are widely used for process optimization. The parameters of this process model were identified by using the historical data from an Australian gold smelting plant. Unfortunately, it is almost prohibitively expensive and time-consuming to build a reliable first principle model in practical industrial processes. Under this situation, Zhang et al.¹¹ proposed to describe the GCLP by a hybrid model, and neural networks were employed to model the kinetic reactions. The developed hybrid model was further utilized in their modifier-adaption (MA) based real-time optimization (RTO) strategy, and the MA method

is used to cope with the plant-model mismatch. Although the reaction rates were estimated by dynamic historical data in their work, only a steady-state optimization problem was formulated to compute the optimal process set-points. To further deal with the disturbances in GCLP, self-optimizing control (SOC) algorithm is adopted by Ye et al.⁵ as a supervisory controller in a two-layer control architecture to maintain the process operations at the optimal set-points. However, the two-layer architecture means that the controller is designed without considering the transient costs explicitly.¹² The operating states of an industrial GCLP fluctuates frequently, and it is impossible to always operate the process around a steady-state set-points. Thus, transient costs should not be ignored in the dynamic operation of GCLP, and the hierarchical separation mentioned above may not necessarily be the economically optimal operation strategy.

Recently, economic model predictive control (EMPC) has been proposed to incorporate a general cost function into the predictive control framework, and could operate a process in a time-varying fashion to optimize the process economics.^{13,14} The closed-loop stability of EMPC methods are also discussed in the literature.^{12–16} Although lots of literature considered the application of classic MPC in the field of hydrometallurgy,^{17–19} few papers apply the EMPC approach to this field. In order to improve the dynamic operation of GCLP, EMPC should be adopted to achieve both transient and steady-state economic optimization simultaneously.²⁰

Another issue of applying EMPC to the GCLP is that there may be a large gap between the nominal process model and the actual plant. In the RTO paradigm, this problem can be addressed by utilizing the last output measurements to modify the steady-state optimization problem.^{2,5,6} For EMPC, the economic performances also strongly rely on the accuracy of the dynamic process model in the resulting optimization problem. However, the classic EMPC may not always guarantee the economic performances due to the plant-model mismatch. Recently, robust nonlinear MPC (NMPC) is proposed to ensure the feasibility of all possible values with tractable computation.²¹ The first version of robust NMPC is min-max MPC, which minimizes the cost of the worst-case realization of the uncertainty while satisfying the constraints for all the cases.^{22,23} Mayne et al.²⁴ extended linear tube-based MPC²⁵ to its nonlinear form. Multi-stage NMPC is proposed by Lucia and Engell²⁶ nowadays, and the uncertainty is perfectly modeled by a scenario tree in their method. Because the multi-stage NMPC calculates the optimal closed-loop feedback policy over a finite prediction horizon, it is suitable for the robust NMPC problem.²⁷ Generally, the uncertain parameters in the process model of GCLP can be divided into two types: The first one can be measured offline, and these historical data can be collected to determine the bounds and distributions of the uncertain parameter; the second type cannot be obtained directly, and they should be estimated by using the prior knowledge and historical data. The bounds and distributions of the uncertain parameters in multi-stage NMPC also need to be determined by known information. And the box uncertainty region used in the

literature may include the unlikely-to-happen scenarios, and further result in the conservative solutions.^{28,29}

Motivated by the above considerations, a multi-stage EMPC is presented in this work to dynamically operate the GCLP under uncertainty. To generate the scenario tree for the proposed multi-stage EMPC, the unmeasurable uncertain parameter in the process model is first estimated from the historical data. To prevent the noise in the measurements from being amplified and propagated into the estimation, a Tikhonov regularization³⁰ based method is used to estimate the reaction rates of GCLP, and the values of the uncertain parameter are computed from the estimated results. Followed by the first estimation step, the bounds of the uncertain parameters in the proposed multi-stage EMPC are determined by kernel density estimation (KDE) method.³¹ The designed uncertainty region could consider the correlations between uncertain parameters, and the possible values (e.g., the minimum, average, and maximum values) of each uncertain parameter can be used to generate the scenarios. Moreover, the probability of each scenario can also be computed from the historical data rather than empirical estimation in multi-stage NMPC. Finally, the states and control inputs are discretized as decision variables to formulate a large nonlinear programming problem (NLP) at each sampling point, and the proposed method is also applied to a simulated industrial GCLP.

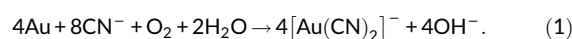
The rest of this work is organized as follows: The next section describes the GCLP and briefly review the multi-stage NMPC. Following the preliminaries, the framework of the proposed multi-stage EMPC is given. Then the estimation results of the uncertain parameter and the strategy for generating the scenario tree are also presented. The mathematical formulation for the multi-stage EMPC of the GCLP is illustrated in this section as well. In the following section, experiments are conducted to verify the efficiency of the proposed method via a simulated GCLP system, and the operating results of the proposed method with different scenario trees are compared. Finally, Conclusions are drawn.

2 | PRELIMINARIES

In this section, the GCLP is first described, and the dynamic process model is built based on mass conservation, followed by the review of the multi-stage NMPC approach.

2.1 | GCLP description

Generally, there are several sub-processes (e.g., ore comminution and grinding, thickening and dewatering, melting and casting, etc.) in a gold smelting plant,² and GCLP is one of the most important stages of the production system. In GCLP, gold is extracted from aerated alkaline slurries by cyanide, and the corresponding overall reaction equation is written as follows



In order to improve the performances of the GCLP, many researchers have investigated its chemistry extensively.⁷⁻⁹ In this work, we focus on a GCLP with several pneumatic CSTRs in a gold smelting plant, as shown in the diagram of Figure 1.

The slurries are fed into the first leaching tank from the buffer tank continuously, and the gold reacts with cyanide ion and oxygen in this leaching tank. The slurries overflow to the subsequent leaching tanks successively, and the same reaction happens in the other leaching tanks. It is beneficial to add all the cyanide in the first leaching tank to reduce the cyanide consumption and speed up the gold dissolution.⁵ If there is insufficient cyanide in the other leaching tanks, it is necessary to consider feeding additional cyanide. To guarantee the leaching rate, which is the ratio of extracted gold component to the overall gold contained in the feeding ore,¹⁰ excessive cyanide is usually added into the leaching tanks. However, when the leaching rate reaches a certain level, it is uneconomical to continue adding cyanide.

After the GCLP, the slurries are further fed into the thickening and dewatering sub-process,³² in which the solid and liquid phases are separated. They should be further processed by the downstream processes to further recover elemental gold. To dynamically operate the GCLP, the first principle model can be used to predict the future behaviors of the process. The dynamic model for the l th ($l = 1, 2, \dots, L$, and L is the number of the leaching tanks) leaching tank consists of the mass conservation of the gold in the solid and liquid phases, and of the cyanide in the liquid phase. The model can be described as follows

$$\frac{dc_{s,l}}{dt} = \frac{q_{s,l}}{m_{s,l}}(c_{s,l-1} - c_{s,l}) - r_{Au,l}. \quad (2)$$

$$\frac{dc_{l,l}}{dt} = \frac{q_{l,l}}{m_{l,l}}(c_{l,l-1} - c_{l,l}) + \frac{m_{s,l}}{m_{l,l}}r_{Au,l}. \quad (3)$$

$$\frac{dc_{CN,l}}{dt} = \frac{q_{l,l}}{m_{l,l}}(c_{CN,l-1} - c_{CN,l}) + \frac{q_{CN,l}}{m_{l,l}} - r_{CN,l}. \quad (4)$$

where $c_{s,l}$ and $c_{l,l}$ are the gold concentration in the solid and liquid phases of the l th leaching tank, respectively, and $c_{CN,l}$ represents the cyanide concentration in the liquid phase of the l th leaching tank. $q_{s,l}$ and $q_{l,l}$ are the feeding flow rate of the solid and liquid phases of the l th leaching tank, respectively. $q_{CN,l}$ is the flow rate of the cyanide solution added into the l th leaching tank, which is the manipulated variable of the GCLP. $r_{Au,l}$ is the gold dissolution rate of the l th leaching tank, and $r_{CN,l}$ is the cyanide consumption rate of the l th leaching tank. $m_{s,l}$ and $m_{l,l}$ are the amounts of the solid and liquid phases in the l th leaching tank, which can be calculated by using the following equations.

$$q_{l,l} = q_{s,l} \left(\frac{1}{c_{w,l}} - 1 \right). \quad (5)$$

$$t_{r,l} = \frac{m_{s,l}}{q_{s,l}} = \frac{m_{l,l}}{q_{l,l}}. \quad (6)$$

where $c_{w,l}$ is the mass concentration of the slurries in the l th leaching tank.

The average residence time of the l th leaching tank $t_{r,l}$ can be computed as follows

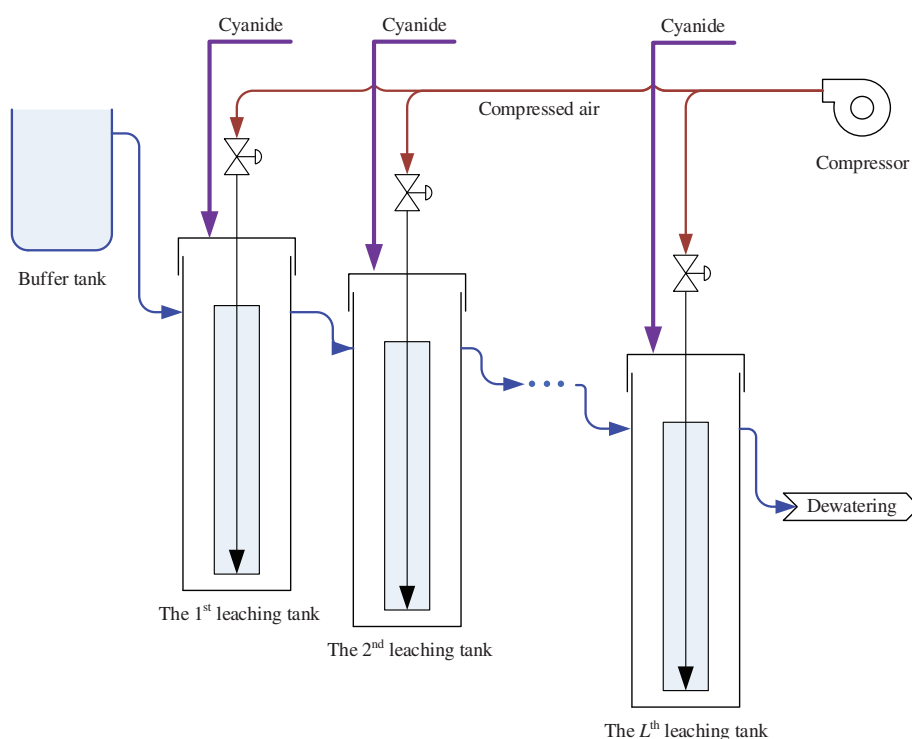


FIGURE 1 The diagram of the GCLP in a gold smelting plant [Color figure can be viewed at wileyonlinelibrary.com]

$$t_{r,l} = \frac{v_l}{q_{s,l}/\rho_s + q_{l,l}/\rho_l}. \quad (7)$$

where v_l is the net reactor volume of the l th leaching tank, and ρ_s and ρ_l are the densities of the solid and liquid phases, respectively.

2.2 | Multi-stage NMPC

The process model used in the multi-stage NMPC method can be described by a set of ordinary differential equations (ODEs) as shown in Equations (2)–(4). A simultaneous implementation of the multi-stage NMPC is usually used to discretize the state and control inputs.^{26,27} Thus, a discretization of the set of ODEs is required firstly. The discretized process model can be formulated as follows

$$\mathbf{x}_{k+1}^j = \mathbf{x}_k^{p(j)} + \Delta t \left(\phi \left(\mathbf{x}_{k+1}^j, \mathbf{u}_k^j, \mathbf{d}_k^{r(j)} \right) \right) \quad (8)$$

where, the states $\mathbf{x}_{k+1}^j$ depend on the parent states $\mathbf{x}_k^{p(j)}$, the corresponding control inputs $\mathbf{u}_k^j$, and the corresponding realization of the uncertainty at the k th sampling point, that is, $\mathbf{d}_k^{r(j)}$. For example, $\mathbf{d}_k^{r(j)} \in \{\mathbf{d}_k^L, \mathbf{d}_k^C, \mathbf{d}_k^U\}$ has three different possible values of uncertainty as shown in Figure 2, where RH means the robust horizon, and PH is for the prediction horizon.²⁷ Since implicit Euler discretization is simple and sufficiently precise,²⁶ it is often used to obtain the discretized model of Equation (8). These discretized states and control inputs are further included as decision variables to formulate the large NLP, and $\mathbf{x}_0$ represents the current initial conditions of the states.

For classic MPC, the control objective is to track the desired set-points.¹⁴ In multi-stage NMPC, a similar optimization objective can also be constructed at each sampling point. In Figure 2, a tree of discretized scenarios in multi-stage NMPC is illustrated, which has the same number of branches at all nodes.^{26,27} The lower and upper bounds of each uncertain parameter are first determined by empirical

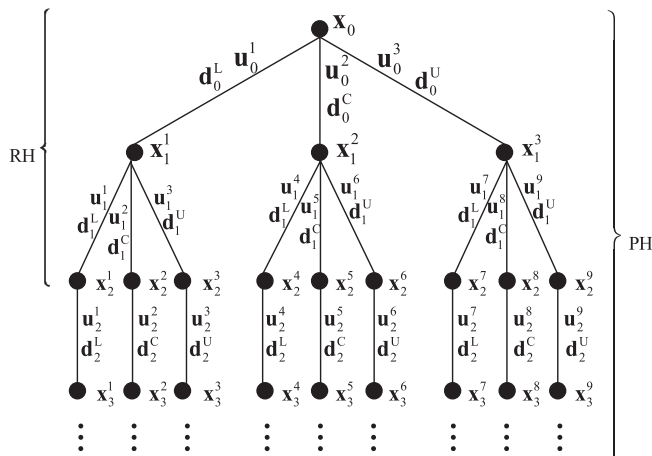


FIGURE 2 Scenario tree of the uncertainty evolution for multi-stage NMPC

estimation, and several equidistant points in this range are used to generate the scenarios, for example, the minimum, average, and maximum values.

If the set of occurring indices (j, k) in the scenario tree is denoted by ξ , the large NLP for multi-stage NMPC can be formulated as follows

$$\begin{aligned} \min_{\mathbf{u}_k^j} \quad & \sum_{i=1}^{N_s} \omega_i \sum_{k=0}^{N_p-1} \left[\left(\mathbf{x}_{k+1}^{(i)} - \mathbf{x}_{sp} \right)^T \mathbf{Q} \left(\mathbf{x}_{k+1}^{(i)} - \mathbf{x}_{sp} \right) \right. \\ & \left. + \left(\mathbf{u}_k^{(i)} - \mathbf{u}_{k-1}^{(i)} \right)^T \mathbf{R} \left(\mathbf{u}_k^{(i)} - \mathbf{u}_{k-1}^{(i)} \right) \right] \\ \text{s.t.} \quad & \mathbf{u}_k^j = \mathbf{u}_k^g \text{ if } \mathbf{x}_k^{p(j)} = \mathbf{x}_k^{p(g)}, \forall (j, k), (g, k) \in \xi \\ & \mathbf{x}_{k+1}^j = \mathbf{x}_k^{p(j)} + \Delta t \left(\phi \left(\mathbf{x}_{k+1}^j, \mathbf{u}_k^j, \mathbf{d}_k^{r(j)} \right) \right), \forall (j, k+1) \in \xi \\ & \mathbf{x}_{k+1}^{(i)} = \mathbf{x}_{k+1}^j, \mathbf{u}_k^{(i)} = \mathbf{u}_k^j \text{ if } \left(\mathbf{x}_{k+1}^j, \mathbf{u}_k^j \right) \in S_i, \forall (j, k), (j, k+1) \in \xi \\ & \mathbf{x}_k^j \in \mathbb{X}, \forall (j, k) \in \xi \\ & \mathbf{u}_k^j \in \mathbb{U}, \forall (j, k) \in \xi \end{aligned} \quad (9)$$

where the i in the superscript bracket represents the states and control inputs corresponding to the i th scenario, ω_i is the probability of the i th scenario, N_s is the number of scenarios, N_p is the prediction horizon, $\mathbf{x}_{sp}$ is the desired set-point of the states, and $\mathbf{Q}$ and $\mathbf{R}$ are the weight matrices corresponding to the states and control inputs, respectively. The first set of the constraints in Equation (9) is the non-anticipativity constraints, and the second set is the discretized process model as shown in Equation (8). $\mathbb{X}$ and $\mathbb{U}$ represent the sets of the states and control inputs, respectively.

3 | MULTI-STAGE EMPC FOR GCLP

To simultaneously achieve both transient and steady-state economic optimization, EMPC could be used to dynamically operate the GCLP. In this section, the framework of the proposed multi-stage EMPC for GCLP is presented that includes two important steps. In the first step, the unmeasurable uncertain parameter in GCLP is estimated by using the Tikhonov regularization based method. In the second step, the scenario tree for multi-stage EMPC is generated based on the estimation results, and the resulting large NLP is formulated to directly optimize the cost function of the GCLP while accounting for the leaching rate constraints at each sampling point.

3.1 | Multi-stage EMPC framework for GCLP

Although the process model of GCLP is described in last section, the semi-empirical models of the reaction rates in Equations (2)–(4) should be further considered. As proposed by the literature,⁴ the nominal gold dissolution rate $r_{Au,l}$ and the cyanide consumption rate $r_{CN,l}$ of the l th leaching tank can be described by the following equations.

$$r_{Au,l} = (a_1 - a_2 d_m^{a_3}) (C_{s,l} - C_{\infty})^{a_4} C_{CN,l}^{a_5} C_{O,l}^{a_6}. \quad (10)$$

$$r_{CN,l} = \left(\frac{b_1}{d_m^{b_2} - b_3} \right) c_{CN,l}^{b_4} \quad (11)$$

where d_m is the average diameter of the ore particles, c_∞ is the residual gold concentration in the solid phase, and $c_{O,l}$ is the dissolved oxygen concentration in the liquid phase of the l th leaching tank that can be measured online. $a_1, a_2, \dots, a_6$ and $b_1, b_2, \dots, b_4$ are the nominal model parameters.

For a GCLP, the average diameter of the ore particles d_m cannot be measured online and has a great effect on the leaching rate. Since d_m often varies due to the fluctuation of the ore comminution and grinding sub-process, using the nominal value of d_m to perform optimal control may not guarantee the economic performances. Another uncertain parameter in GCLP is the residual gold concentration in the solid phase c_∞ , which can be determined by d_m and other properties of the ore, and it cannot be measured directly. Although the empirical model can be employed to describe the correlations between c_∞ and d_m , it is impossible to gather such data online for process optimization in the actual GCLP. Moreover, the identified semi-empirical model may not predict the GCLP adequately due to the influences of the other unknown factors.

To this end, the multi-stage EMPC can be used to cope with the effects of the uncertain parameters in the process model. For the

average diameter of ore particles, the history data can be used to determine its distribution, which could be further used to generate the scenario tree in the proposed multi-stage EMPC. However, for uncertain parameter c_∞ , direct measurement is intractable. Therefore, regularization methods are required to estimate its bounds and distribution. Because of the above considerations, the framework of the proposed multi-stage EMPC for GCLP is illustrated in Figure 3. The bounds and distributions of the uncertain parameters are first estimated for generating the scenario tree. Based on the results obtained from the first step, the NLP for multi-stage EMPC is formulated, and the receding horizon control optimizes the cost function over a prediction horizon while accounting for the leaching rate constraints.

Remark 1 Although the values of the average diameter of the ore particles can be measured by laboratory analysis, the continuous measurements cannot be obtained accurately in an industrial GCLP. To this end, the screen mesh is often used to fast measure the approximate values of this uncertain parameter. Thus, the discrete probability distribution for the average diameter of the ore particles is used to estimate the uncertainty region in this work, and the distribution can be obtained by computing the frequency of occurrence in the historical data. On the other

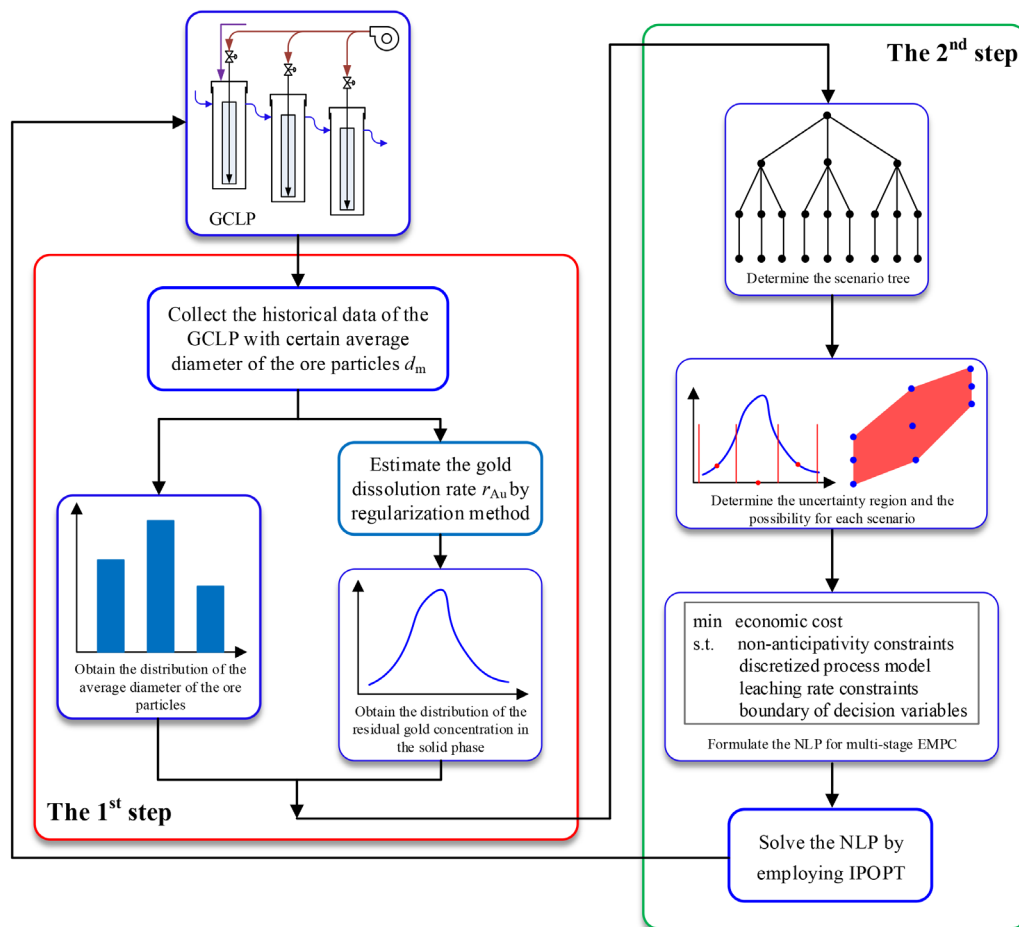


FIGURE 3 The proposed multi-stage EMPC framework for GCLP [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

hand, the residual gold concentration in the solid phase generally hard to measure in actual GCLP. Since Tikhonov regularization based method is used to estimate the values of this uncertain parameter from the continuous input and output measurements, continuous estimations can be obtained. Therefore, the continuous probability distribution for the residual gold concentration in the solid phase is considered in this work, and the distribution can be achieved by using the KDE method. These results can be employed to further generate the scenario tree in the following section.

3.2 | Estimating the unmeasurable uncertain parameter

In order to generate the scenario tree for multi-stage EMPC, the unmeasurable uncertain parameter c_∞ should be estimated first. In multi-stage NMPC,^{26,27} the uncertain information of the parameters is determined by empirical estimation. However, the box uncertainty region may contain the unlikely-to-happen scenarios, and the solution of the optimization problem (9) may be too conservative. Since the uncertain parameter c_∞ cannot be directly measured, the gold dissolution rate $r_{Au,l}$ in Equation (2) is required to be estimated first. And then the nonlinear Equation (10) can be solved to obtain the values of c_∞ with a certain value of d_m . It is apparent that estimating the reaction rate by Equation (2) needs to compute the derivatives of the measurements of the gold concentration in the solid phase with respect to time, and it is usually an ill-posed inverse problem.³⁰ Thus, proper estimation methods are required to tackle the noise in the concentration measurements from being amplified and propagated into the estimation of the reaction rates. To solve the ill-posed inverse problems, a number of approaches have been proposed.^{33,34} In this work, a Tikhonov regularization based estimation method³⁰ is used, and such a method can reduce the effects of the measurement noise as much as possible.

To estimate the reaction rates accurately, the dynamic process model should be transformed. If the gold concentration in the solid phase is obtained, Equation (2) can be used to estimate the reaction rate $r_{Au,l}$, which can be further described by the following equation.

$$y_{s,l}(t) = \frac{q_{s,l}}{m_{s,l}} \int_{\tau=0}^t [c_{s,l-1}(\tau) - c_{s,l}(\tau)] d\tau - c_{s,l}(t) + c_{s,l}(0). \quad (12)$$

where $y_{s,l}$ represents the generalized gold concentration in the solid phase of the l th leaching tank. The original reaction rate $r_{Au,l}$ can then be computed as follows

$$r_{Au,l} = \frac{dy_{s,l}(t)}{dt}. \quad (13)$$

Similar results can also be obtained from Equation (3). Since c_∞ is the only unmeasurable uncertain parameter, either Equation (3) or (4)

can be employed to estimate the reaction rate $r_{Au,l}$. By introducing the generalized gold concentration in the liquid phase of the l th leaching tank, Equation (13) can be further rewritten in the following form.

$$y_{s,l}(t) = \int_0^t r_{Au,l}(\tau) d\tau + y_{s,l}^0. \quad (14)$$

where $y_{s,l}^0$ is the initial generalized gold concentration.

If inappropriate methods are used to estimate $r_{Au,l}$, the noise of the generalized gold concentration in the solid phase $y_{s,l}$ will be amplified, and leading to inaccurate estimation. Thus, instead of calculating $r_{Au,l}$ in Equation (13) directly, Equation (14) is further reformulated by integration by parts as follows

$$\begin{aligned} y_{s,l}(t) &= \int_{\tau=0}^t \left\{ \frac{d[\tau r_{Au,l}(\tau)]}{d\tau} - \tau f_{s,l}(\tau) \right\} d\tau + y_{s,l}^0 \\ &= \tau r_{Au,l}(\tau) \Big|_{\tau=0}^t - \int_{\tau=0}^t \tau f_{s,l}(\tau) d\tau + y_{s,l}^0 \\ &= \tau r_{Au,l}(t) - \int_{\tau=0}^t \tau f_{s,l}(\tau) d\tau + y_{s,l}^0 \\ &= t \left[\int_{\tau=0}^t f_{s,l}(\tau) d\tau + r_{Au,l}^0 \right] - \int_{\tau=0}^t \tau f_{s,l}(\tau) d\tau + y_{s,l}^0. \end{aligned} \quad (15)$$

where $f_{s,l}(t) = dr_{Au,l}(t)/dt$, and $r_{Au,l}^0$ is the initial reaction rate of the l th leaching tank. After rearranging Equation (15), the following equation can be obtained.

$$y_{s,l}(t) = \int_{\tau=0}^t (t-\tau) f_{s,l}(\tau) d\tau + y_{s,l}^0 + r_{Au,l}^0. \quad (16)$$

If $f_{s,l}(t)$, $y_{s,l}^0$, and $r_{Au,l}^0$ are computed, the values of $r_{Au,l}(t)$ can be obtained by direct numerical integration.

By discretizing Equation (16), the following equation can be obtained.

$$\hat{y}_{s,l}^m = \sum_{t_n=0}^{t_n=t_m} \alpha_{mn} (t_m - t_n) f_{s,l}^n \Delta t + y_{s,l}^0 + r_{Au,l}^0. \quad (17)$$

where $\hat{y}_{s,l}$ represents the generalized gold concentration in the solid phase of the l th leaching tank predicted by Equation (17). The superscript $m = 1, 2, \dots, M$, and M is the total number of samples. The final time t_f is divided into N uniformly spaced discretization points with step $\Delta t = t_f/(N-1)$, and $n = 1, 2, \dots, N$. $f_{s,l}^1, f_{s,l}^2, \dots, f_{s,l}^N$ are the discretized $f_{s,l}(t)$, and α_{mn} is determined by using Simpson's 1/3 rule.³⁰ When all of the data points are considered, the vector of the prediction errors obtained from Equation (17) can be rewritten in the following matrix form.

$$\mathbf{e}_{s,l} = \tilde{\mathbf{Y}}_{s,l} - \mathbf{A} \mathbf{f}_{s,l} - \mathbf{1}_M y_{s,l}^0 - \mathbf{t}_M r_{Au,l}^0. \quad (18)$$

where $\tilde{\mathbf{y}}_{s,l}$ is the vector of the actual generalized gold concentration in the solid phase of the l th leaching tank. $\mathbf{A}$ is a $M \times N$ matrix of the coefficients of the vector $\mathbf{f}_{s,l}$, of which the elements are α_{mn} in

Equation (18), and $\mathbf{f}_{s,l} = [f_{s,l}^1, f_{s,l}^2, \dots, f_{s,l}^N]^T$. $\mathbf{1}_M$ is the $M \times 1$ column vector of one, $\mathbf{t} = [t_1, t_2, \dots, t_M]^T$, and $t_M = t_f$.

Due to the noise in the historical data, minimizing $\mathbf{e}_{s,l}^T \mathbf{e}_{s,l}$ will not obtain a smooth $f_{s,l}(t)$. To ensure the smoothness, additional conditions have to be imposed. In Tikhonov regularization,³⁵ the minimization of the sum of squares of the second-order derivation $d^2 f_{s,l}/dt^2$ at the discretization points can be used. In terms of the vector $\mathbf{f}_{s,l}$, the additional condition can be calculated by the following equation.

$$\sum_{n=2}^{N-1} \left(d^2 f_{s,l}^n / dt^2 \right)^2 = (\boldsymbol{\beta} \mathbf{f}_{s,l})^T (\boldsymbol{\beta} \mathbf{f}_{s,l}). \quad (19)$$

where the tri-diagonal coefficient matrix $\boldsymbol{\beta}$ can be obtained from the finite difference approximation as follows

$$\boldsymbol{\beta} = \frac{1}{\Delta t^2} \begin{bmatrix} 1 & -2 & 1 & & & \\ & 1 & -2 & 1 & & 0 \\ & & \dots & \dots & \dots & \\ 0 & & & 1 & -2 & 1 \\ & & & & 1 & -2 & 1 \end{bmatrix}. \quad (20)$$

Instead of minimizing $\mathbf{e}_{s,l}^T \mathbf{e}_{s,l}$ and $d^2 f_{s,l}/dt^2$ separately, a linear combination of these two terms is optimized. Thus, $\mathbf{f}_{s,l}$ can be computed by solving the following optimization problem.

$$\min \mathbf{e}_{s,l}^T \mathbf{e}_{s,l} + \lambda \mathbf{f}_{s,l}^T \boldsymbol{\beta}^T \boldsymbol{\beta} \mathbf{f}_{s,l}. \quad (21)$$

where λ is an adjustable regularization factor, which is determined by generalized cross-validation³⁰ in this work.

By minimizing the objective function (21), the linear algebraic solution can be obtained as follows

$$\tilde{\mathbf{f}}_{s,l} = (\tilde{\mathbf{A}}^T \tilde{\mathbf{A}} + \lambda \tilde{\boldsymbol{\beta}}^T \tilde{\boldsymbol{\beta}})^{-1} \tilde{\mathbf{A}}^T \tilde{\mathbf{y}}_{s,l}. \quad (22)$$

where $\tilde{\mathbf{f}}_{s,l} = [\mathbf{f}_{s,l}^T, y_{s,l}^0, r_{Au,l}^0]^T$, $\tilde{\mathbf{A}} = [\mathbf{A}, \mathbf{1}_M, \mathbf{t}]$, $\tilde{\boldsymbol{\beta}} = [\boldsymbol{\beta}, \mathbf{0}_{(N-2) \times 2}]$, and the vector $\mathbf{0}_{(N-2) \times 2}$ represents the $(N-2) \times 2$ matrix of zero. Once the optimal solution $\tilde{\mathbf{f}}_{s,l}^*$ is obtained, the reaction rate $r_{Au,l}$ can be estimated by using the following integration.

$$r_{Au,l}(t) = \int_{\tau=0}^t f_{s,l}(\tau) d\tau. \quad (23)$$

Although Equation (10) is a nonlinear equation, the empirical range of the uncertain parameter c_∞ can be easily determined by the prior knowledge, that is, the parameter value is always less than the gold concentration in the solid phase that flows out of the l th leaching tank and greater than zero. Thus, the nonlinear Equation (10) can be solved to obtain the only solution of c_∞ with the given value of d_m .

Remark 2 In order to estimate the derivatives of the concentration measurements with respect to time, several methods have

been presented in the literature.^{2,3,33,34} Since the estimation is an ill-posed inverse problem, a smooth function can be fitted to the data, and the derivatives are computed from the fitted function. The most commonly used method is the finite difference method, where the current sample, as well as several nearby samples, are used to approximate the current derivative.^{2,3} To avoid the amplification and propagation of the noise, smoothing techniques were developed to compute the derivative estimates from the noisy data accurately. One of the most widely used method is polynomial fitting, in which a smooth polynomial function is fitted to the historical data, and the fitted polynomial function is differentiated analytically to obtain the derivatives.^{33,34} Recently, a new method for processing the time-concentration data of the reaction kinetics is proposed by Yeow et al.³⁰ Since the Tikhonov regularization and generalized cross-validation approaches are used in their method, the amplification and propagation of the noise can be kept under control more efficiently. In this work, the Tikhonov regularization based estimation method is also employed, because it is easy to implement in an industrial GCLP, and can further reduce the negative effects of noise at the same time. This is an important area for future research, but it is out of the scope of this work.

3.3 | Kernel density estimator

A kernel distribution is a nonparametric representation of the probability density function (pdf) of a random variable, which is defined by a kernel smoothing function $\kappa(\cdot)$ and a bandwidth w . For uncertain parameter c_∞ , the Kernel density estimator (KDE) method can be described by using the following equation.³¹

$$\hat{g}_{\text{KDE}}(c_\infty) = \frac{1}{Mw} \sum_{m=1}^M \kappa\left(\frac{c_\infty - c_{\infty,m}}{w}\right). \quad (24)$$

where $c_{\infty,1}, c_{\infty,2}, \dots, c_{\infty,M}$ are random samples from the historical dataset of c_∞ , and M is also the number of available samples. In the KDE method, $\kappa(\cdot)$ defines the shape of the curve used to generate the pdf, and w is used to control the smoothness of the resulting density curve. The kernel estimator for the cumulative distribution function (cdf) can be calculated as follows

$$\hat{G}_{\text{KDE}}(c_\infty) = \int_{-\infty}^{c_\infty} \hat{g}_{\text{KDE}}(\tau) d\tau. \quad (25)$$

To avoid making assumptions about the distribution of the residual gold concentration in the solid phase, the KDE method is employed to estimate its pdf first. After obtaining the estimation results, the quantile function can be used to acquire the confidence interval of the residual gold concentration in the solid phase for a specific confidence level, which is defined by using the following equation.

$$\hat{G}_{KDE}^{-1}(\gamma) = \min\{c_{\infty} \in \zeta | \hat{G}_{KDE}(c_{\infty}) \geq \gamma\}. \quad (26)$$

where ζ is the set of c_{∞} , and γ is the specific confidence level. Thus, the lower and upper bounds for c_{∞} are $\hat{G}_{KDE}^{-1}(\gamma)$ and $\hat{G}_{KDE}^{-1}(1-\gamma)$, respectively.

If the confidence level γ increases, the confidence interval will become smaller. Under this situation, the generated scenario tree may not include all the possible realizations of the uncertainty, and might lead to infeasibility of the optimization problem. On the contrary, a larger confidence interval that contains outliers would make the control policies too conservative.

3.4 | Generating the scenario tree for multi-stage EMPC

The first step for multi-stage EMPC is to generate the scenario tree as in multi-stage NMPC.²⁷ For multi-stage EMPC, the RH is also the branching of the scenario tree up to a certain stage, based on the assumption that the uncertainty remains constant until the end of the PH. Figure 2 corresponds to the scenario tree with RH equal to 2. The major challenge of multi-stage EMPC is to solve the resulting large NLP in a reasonable amount of time. Thus, to avoid the exponential growth of the scenario tree, the proper value of RH should be determined. Since the far future does not have to be represented as exactly as the next stage, and the states and control inputs will be refined later, the scenario tree can be branched only in the first stage.²⁷ Another important issue is to determine the uncertainty region of GCLP. If the bounds of the uncertain parameters are too large, the solution will be more conservative and result in economic loss. On the contrary, the extreme values that are not included in the scenario tree may violate the constraints. To this end, it is necessary to define the scenarios properly.

Assuming that the average diameter of the ore particles would not vary in a short period, and several possible values can be obtained for uncertain parameter d_m , for example, d_m^L , d_m^C , and d_m^U for the minimum, average and maximum values, respectively. For maximum possible value d_m^U , M_U available samples in the historical dataset are collected to estimate the uncertain residual gold concentration in the solid phase, and $\hat{c}_{\infty,m}^U$ ($m = 1, 2, \dots, M_U$) is used to represent the m th estimated value of the residual gold concentration in the solid phase under the possible value d_m^U . Under this situation, the results of KDE can be used to estimate the bounds of the uncertain parameter c_{∞}^U , which can be written as follows

$$\hat{g}_{KDE}^U(c_{\infty}^U) = \frac{1}{M_U} \sum_{m=1}^{M_U} \kappa_G(c_{\infty}^U, \hat{c}_{\infty,m}^U). \quad (27)$$

In this work, Gaussian kernel function is used, since it is widely employed in kernel smoothing methods.²⁸ And the Gaussian kernel can be described by the following equation.

$$\kappa_G(c_{\infty}^U, \hat{c}_{\infty,m}^U) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\|c_{\infty}^U - \hat{c}_{\infty,m}^U\|^2}{2\sigma^2}\right). \quad (28)$$

where σ is the bandwidth parameter of Gaussian kernel function. If $\hat{G}_{KDE}^U(c_{\infty}^U)$ is the cdf of c_{∞}^U , the lower and upper bounds of c_{∞}^U can be defined by $c_{\infty,L}^U = \hat{G}_{KDE}^{U-1}(\gamma)$ and $c_{\infty,U}^U = \hat{G}_{KDE}^{U-1}(1-\gamma)$, respectively.

Another parameter that also plays an important role in multi-stage EMPC is the probability of the i th scenario ω_i in Equation (9), which can be estimated by using the results from KDE as well. Since the average diameter of the ore particles is directly measured by screen mesh, the history data can be used to estimate the probability of d_m^U easily. Assuming that the probability of d_m^U is defined as $p(d_m^U)$, the probability of the seventh scenario ω_7 in Equation (9) can be calculated by $\omega_7 = p(d_m^U) \cdot p(c_{\infty,L}^U)$. In order to determine $p(c_{\infty,L}^U)$ in this work, the calculated range of c_{∞}^U is divided into two equal intervals firstly. If we also consider three possible values for c_{∞}^U , the central point (average) can be easily computed as $c_{\infty,C}^U = (c_{\infty,L}^U + c_{\infty,U}^U)/2$. By defining $\delta^U = (c_{\infty,U}^U - c_{\infty,L}^U)/4$, the probability of $c_{\infty,L}^U$ can be calculated as follows

$$\tilde{p}(c_{\infty,L}^U) = \int_{c_{\infty,L}^U - \delta^U}^{c_{\infty,L}^U + \delta^U} \hat{g}_{KDE}^U(\tau) d\tau. \quad (29)$$

Similarly, $\tilde{p}(c_{\infty,C}^U)$ and $\tilde{p}(c_{\infty,U}^U)$ can be obtained by using the same way.

In this work, since the quantile function used to determine the confidence interval (the lower and upper bounds) of the uncertainty parameter c_{∞} for a specific confidence level, the sum of the probabilities of the scenarios under a certain average diameter of the ore particles may not always equal to one, that is, $\tilde{p}(c_{\infty,L}^U) + \tilde{p}(c_{\infty,C}^U) + \tilde{p}(c_{\infty,U}^U) \neq 1$. To make the sum of the probabilities remaining equal to one, $\tilde{p}(c_{\infty,L}^U)$ obtained from Equation (29) should be further normalized by using the following equation.

$$p(c_{\infty,L}^U) = \frac{\tilde{p}(c_{\infty,L}^U)}{\tilde{p}(c_{\infty,L}^U) + \tilde{p}(c_{\infty,C}^U) + \tilde{p}(c_{\infty,U}^U)}. \quad (30)$$

And the same strategy can be applied to obtain the other probabilities of the scenarios in Equation (9) as well.

Remark 3 In this work, the uncertainty of GCLP comes from the feed slurries, and the scenario tree in the proposed method is generated by using the historical data. Generally, the average diameter of the ore particles in the slurries is determined by the properties of the ore, as well as the ore comminution and grinding sub-process. The residual gold concentration in the solid phase is not only related to the average diameter of the ore particles but also influenced by the properties of the ore. Since there is no equipment to alter the average diameter of the ore particles in GCLP, the distributions of these two uncertain parameters will remain the same for any operating

strategies, and the corresponding scenario probabilities are constants as well. In other words, the values of the two uncertain parameters would not be affected by the amount of cyanide that is added into the leaching tanks. Consequently, the average diameter of the ore particles and the residual gold concentration in the solid phase are time-independent parameters in GCLP. Thus, the historical data of the uncertain parameters gathered under the non-EMPC method can still be applied to determine the uncertainty set of the proposed multi-stage EMPC approach. Because it is difficult to quantitatively describe the correlation between the two parameters, the scenario tree is appropriate to capture the uncertainty relationship.

3.5 | Mathematical formulation of multi-stage EMPC

After generating the scenario tree, the objective function of the multi-stage EMPC for GCLP should be defined first. Since the dissolved oxygen in the slurries of each leaching tank is provided by air compressor as shown in Figure 1, the corresponding energy consumption is almost constant. Thus, the cost of the air compressor is not included in the objective function. Under the condition of low cyanide consumption and fast gold dissolution, all cyanide is required to be added into the first tank; otherwise, it can be added to other tanks to increase the leaching rate. Since the proposed multi-stage EMPC has strong robustness under uncertainty, the cyanide is considered only adding to the first tank in this work. The objective of EMPC for GCLP is to reduce the production cost, which is generally composed of three parts: the cyanide consumption cost, the cost for disposing of the residual cyanide, and the cost of gold outflowing from the L th leaching tank. To this end, the whole production cost for classic EMPC can be formulated as follows

$$J = \sum_{k=0}^{N_p-1} (P_{CN} q_{CN,1k} + P_{dCN} q_{iLk} c_{CN,Lk} + P_{dAu} q_{sLk} c_{s,Lk}). \quad (31)$$

where P_{CN} is the price of the cyanide solution, and P_{dCN} and P_{dAu} are the price for disposing the residual cyanide and gold, respectively. The subscript “ k ” here refers to the k th sampling point.

Another important performance index in GCLP is the leaching rate. To make the whole gold smelting plant operate smoothly, the leaching rate of the GCLP should not be less than the predesigned value. Thus, the process constraints in GCLP can be written by the following equation.

$$\hat{a}_k = \frac{c_{s,0} - c_{s,Lk}}{c_{s,0}} \times 100\% \geq \bar{a}. \quad (32)$$

where $c_{s,0}$ is the gold concentration in the solid phase that is fed into the first leaching tank, $\hat{a}_k$ is the calculated leaching rate at the k th sampling point, and $\bar{a}$ is the predesigned leaching rate.

In this work, the implicit Euler discretization is also used to discretize the process model in Equations (2)–(4), and the discretized states and control inputs are adopted as the decision variables in the resulting NLP, which can be constructed as follows

$$\mathbf{c}^{NLP} = [c_0^T, c_1^T, c_1^{2T}, \dots, c_{N_p}^{N_p T}, q_{CN,10}^1, q_{CN,10}^2, \dots, q_{CN,1N_p-1}^{N_p}]^T \quad (33)$$

where $\mathbf{c}_0 = [c_{s,10}, c_{i,10}, c_{CN,10}, \dots, c_{s,L0}, c_{i,L0}, c_{CN,L0}]^T$, $c_{s,10}$ is the initial gold concentration in the solid phase of the l th leaching tank, and $c_{i,10}$ and $c_{CN,10}$ are the initial gold and cyanide concentration in liquid phase of the l th leaching tank, respectively. Similar to the initial condition vector $\mathbf{c}_0$, the k th state vector is $\mathbf{c}_k^j = [c_{s,1k}^j, c_{i,1k}^j, c_{CN,1k}^j, \dots, c_{s,Lk}^j, c_{i,Lk}^j, c_{CN,Lk}^j]^T$. Since the set of occurring indices (j, k) in the scenario tree is denoted by ξ , the resulting NLP for the proposed multi-stage EMPC at each sampling point can be formulated as follows

$$\begin{aligned} \min_{\mathbf{c}^{NLP}} \quad & \sum_{i=1}^{N_k} \omega_i \sum_{k=0}^{N_p-1} (P_{CN} q_{CN,1k}^{(i)} + P_{dCN} q_{iLk}^{(i)} c_{CN,Lk}^{(i)} + P_{dAu} q_{sLk}^{(i)} c_{s,Lk}^{(i)}) \\ \text{s.t.} \quad & q_{CN,1k}^j = q_{CN,1k}^g \text{ if } \mathbf{c}_k^{p(j)} = \mathbf{c}_k^{p(g)}, \forall (j,k), (g,k) \in \xi \\ & \mathbf{c}_{k+1}^j = \mathbf{c}_k^{p(j)} + \Delta t \left(\tilde{\phi} \left(\mathbf{c}_{k+1}^j, q_{CN,k}^j, \mathbf{d}_k^{r(j)} \right) \right), \forall (j,k+1) \in \xi \\ & \mathbf{c}_{k+1}^{(i)} = \mathbf{c}_{k+1}^j, q_{CN,k}^{(i)} = q_{CN,k}^j, \text{ if } \left(\mathbf{c}_{k+1}^j, q_{CN,k}^j \right) \in S_i, \forall (j,k), (j,k+1) \in \xi \\ & \hat{a}_k \geq \bar{a}, \forall (j,k) \in \xi \\ & \mathbf{c}_L \leq \mathbf{c}^{NLP} \leq \mathbf{c}_U. \end{aligned} \quad (34)$$

where $\tilde{\phi}(\cdot)$ represents the discretized process model in Equations (2)–(4), $\mathbf{c}_L$ and $\mathbf{c}_U$ are the lower and upper bounds of $\mathbf{c}^{NLP}$, respectively. S_i is also the set of the i th scenario. The uncertain parameters are $\mathbf{d} = [d_m, c_{\infty}]^T$. Different from the objective in Equation (9), the optimization problem (34) accounts directly for the process economics in the proposed multi-stage EMPC.

Remark 4 The major computational challenge for multi-stage EMPC is to solve the resulting large NLP in a reasonable time. The size of the NLP increases exponentially with the control horizon, the number of uncertain parameters, and the uncertainty levels in the designed scenario tree. Although the issue has been studied in stochastic programming extensively, they are only suitable for convex multi-stage optimization problem.²⁷ On one hand, reducing the growth of the NLP problem size can be achieved by considering the specific tree branches up to RH as in multi-stage NMPC.²⁶ On the other hand, for an industrial GCLP, the uncertain parameters can be determined by using the prior knowledge, and the historical data can be used to further generate a scenario tree with a good representation of the uncertainty in GCLP. In order to obtain the solution of the NLP problem in the sampling interval, the minimum, average, and maximum values of each uncertain parameter are considered to generate nine scenarios in this work, which is supported by simulation results in the following section.

4 | APPLICATION TO A SIMULATED INDUSTRIAL GCLP

This section presents a comparison of the economic performances of the classic EMPC and the multi-stage EMPC methods with different scenario trees for a simulated industrial GCLP. For these EMPC methods compared in this work, a finite horizon optimal control problem is solved to obtain the optimal control sequence at each sampling point, and the initial states are selected as the current states of the GCLP. The first control action in this sequence is applied to the GCLP, while at the next sampling point the whole procedure is repeated, following a rolling horizon implementation. To generate the scenario tree by using the historical data, the uncertain parameter estimation method is first evaluated. Three case studies are then implemented to evaluate the economic performance of the proposed multi-stage EMPC approach. The resulting NLPs are solved by employing IPOPT via MATLAB/OPTI toolbox. The hardware used in this work is a 4-core Intel i5 processor (1.60 GHz) with 8 GB of RAM.

4.1 | A simulated industrial GCLP description and data generation

The GCLP studied in this work consists of four equally-sized pneumatic leaching tanks that connect each other by overflow. The GCLP is part of a gold smelting plant as shown in Figure 4. For an industrial GCLP, the feeding flow rate and concentration of the slurries can be measured by an electromagnetic flowmeter and online densitometer, and the dissolved oxygen analyzer takes the oxygen concentration in

the liquid phase every second. The gold and cyanide concentration in the liquid phase can be measured by an automatic titration analyzer, and the gold concentration in the solid phase can be taken by an X-ray fluorescence analyzer. The cyanide solution is added by the peristaltic pump, which is the manipulated variable in this GCLP. According to the actual process mentioned above, the first principle model of the GCLP is first built to simulate the actual process. The equipment parameters in the first principle model are consistent with the actual GCLP, and the process data are used to identify the parameters in Equations (10)–(11). Based on the first principle models of the GCLP and other sub-processes, a “plant-wide simulation platform for gold hydrometallurgy” is developed, and the virtual automation system is also established to make the simulation more realistic. The simulation platform in Figure 4 is used to illustrate the efficiency of the proposed multi-stage EMPC method for GCLP in this work. Therefore, the experimental results may be similar to the actual industrial process performance, although the experiments are performed through the simulation platform.

The empirical model used in this work to describe the residual gold concentration in the solid phase can be written as follows

$$c_{\infty}(d_m) = c_1 [1 - c_2 \exp(c_3 d_m)] + e_m. \quad (35)$$

where c_1 , c_2 , c_3 are the nominal parameters of the empirical model, and e_m represents the unknown factors that affect the residual gold concentration in the solid phase. Since there are correlations between d_m and c_{∞} , for classic EMPC, Equations (2)–(4) and Equations (10)–(11) should be integrated with Equation (35) to predict the future behaviors of the GCLP accurately. However, for an industrial GCLP, it

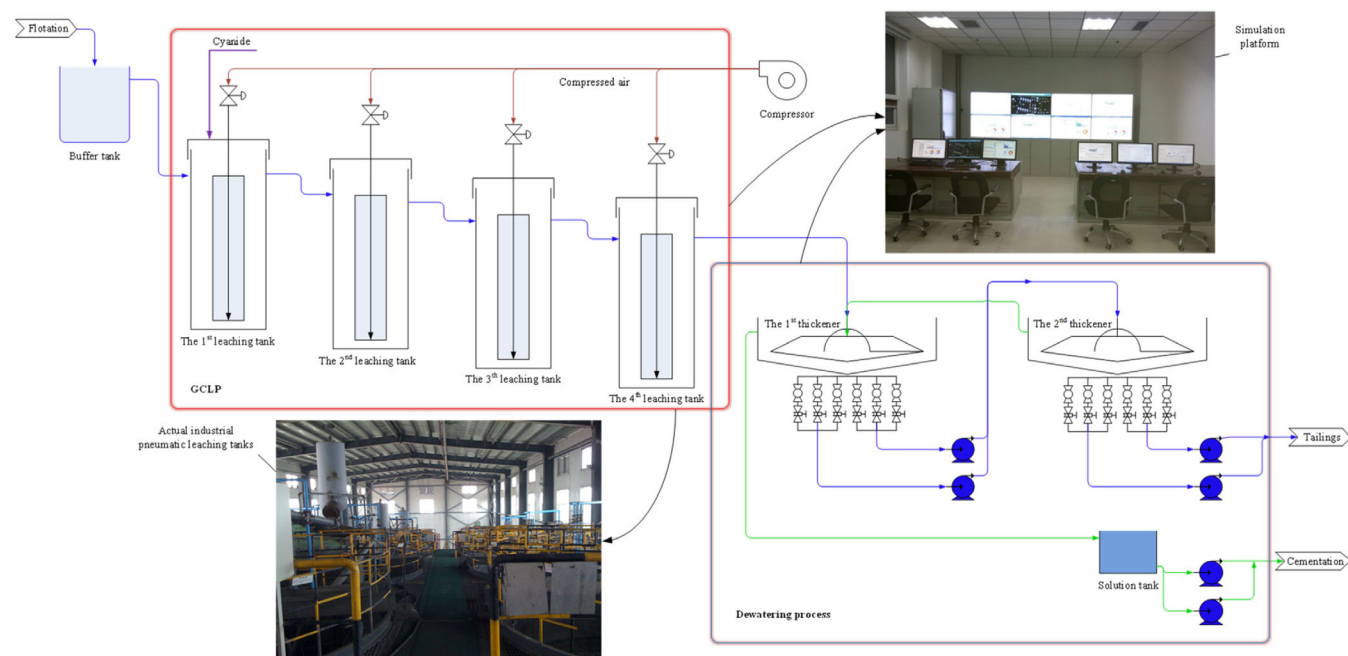


FIGURE 4 The flowsheet of the industrial GCLP in a gold hydrometallurgy plant and its simulation platform [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

is difficult to build the empirical model (35), and the established empirical model could be mismatched with the actual process as well. For the proposed multi-stage EMPC method, the empirical model (35) is not required to formulate the large NLP (34).

In order to generate the historical data for estimating the uncertain parameter c_{∞} , the average diameter of the ore particles d_m is first assumed to have three possible values in this work, that is, $d_m = \{60, 80, 100\}$. Under each possible value of d_m , 100 samples of the residual gold concentration in the solid phase are generated according to Equation (35), and the unknown factor is normally distributed, that is, $e_m \sim N(0, 0.02)$. Equations (2)–(4) are integrated with Equations (10)–(11) and Equation (35) to generate the historical data for estimating the uncertain parameter c_{∞} , and 5% noise is also added into the measurements of the states in Equations (2)–(4) as used in the literature.² The nominal parameters used in this work are illustrated in Table 1.

4.2 | Estimating the residual gold concentration in the solid phase

In this work, the four equal-sized pneumatic leaching tanks are connected to each other by overflow, so the feeding flow rate and concentration of the slurries, as well as the net reactor volume for each leaching tank, are considered to be the same. Since the same ore flows from one leaching tank to another, we can also assume that the four tanks have the same reaction rates in a short period, and any tank can be chosen to estimate the uncertain residual gold concentration in the solid phase. The cheapest way is to use the first leaching tank, because the initial gold concentration in the ore $c_{s,0}$ would keep constant in this short period. Under this situation, Equation (12) can be simplified as follows

$$y_{s,1}(t) = \frac{q_s}{m_s} \left[c_{s,0}t - \int_0^t c_{s,1}(\tau) d\tau \right] - c_{s,1}(t) + c_{s,1}(0) \quad (36)$$

where q_s and m_s are the flow rate and amount of the solid phase in each leaching tank, respectively. $c_{s,1}(0)$ represents the initial gold concentration of the solid phase in the first leaching tank.

The estimated results for the gold dissolution rate r_{Au} are first drawn in Figure 5. In this figure, 100 samples of the gold concentration in the solid phase are used to forecast r_{Au} with the condition of

$d_m = 80 \mu\text{m}$. The adjustable regularization factor λ in Equation (21) is determined by generalized cross-validation,³⁰ and the optimal value of λ is 10^4 . In Figure 5, the Tikhonov regularization based estimation method can handle the noise in the measurements and estimate the trend of the actual gold dissolution rate accurately.

In Table 2, the finite difference method and polynomial fitting method are also used to estimate the gold dissolution rate. The root mean square error (RMSE) and the mean absolute error (MAE) indices are employed to evaluate the estimation performances. Since the finite difference method cannot deal with the noise in the historical data, the worst case estimating results are obtained. When the polynomial fitting method is used to estimate the derivatives, the results are improved. This is because the polynomial fitting method can suppress the influence of noise to a certain extent. The best estimating results are obtained by the Tikhonov regularization based method, since the noise in the historical data is well handled by solving the optimization problem (21). Thus, this method is used to estimate the gold dissolution rate r_{Au} in this work.

To further evaluate the efficiency of the estimation method, the estimated results with the other possible values of d_m are also illustrated in Table 3. When d_m is equal to $80 \mu\text{m}$, the RMSE is 0.0127, and the MAE is 0.0417. If the historical data with $d_m = 60 \mu\text{m}$ or $d_m = 100 \mu\text{m}$ is used to forecast the gold dissolution rate, similar estimation performances can be achieved, as shown in Table 3.

After solving the nonlinear Equation (10), the estimated value of the residual gold concentration in the solid phase can also be obtained. As illustrated by the estimation results in Table 3 the proposed method estimates the uncertain parameter c_{∞} precisely. The RMSEs are about 0.0070, and the MAEs are around 0.0300. These results can be used to further generate the scenario tree in the next section.

Finally, by applying KDE method, the pdfs are compared in Figure 6. The blue solid lines represent the pdfs estimated by using the actual values of c_{∞} , and the red dotted lines are the pdfs identified by using the estimated values $\hat{c}_{\infty}$. In this figure, the estimated pdfs of c_{∞} and $\hat{c}_{\infty}$ with the same d_m are quite similar, so the proposed method can efficiently determine the distribution of the uncertain parameter c_{∞} in this case.

4.3 | Generating the scenario tree

For multi-stage EMPC, the bounds and distribution of c_{∞} with different possible values of d_m is required to generate the scenario tree.

TABLE 1 The nominal parameter values in the process model of the GCLP as the simulated reality

Parameter	Nom. value	Parameter	Nom. value	Parameter	Nom. value
q_s	2,500 kg/hr	P_{CN}	\$2.11/kg	a_6	0.228
c_w	39%	P_{dCN}	\$0.35/kg	b_1	1.69×10^{-8}
$c_{s,0}$	20 mg/kg	P_{dAu}	\$31.83/g	b_2	0.547
d_m	80 μm	a_1	1.13×10^{-3}	b_3	6.40
v	68 m^3	a_2	4.37×10^{-11}	b_4	3.71
ρ_s	2,800 kg/ m^3	a_3	2.93	c_1	0.357
ρ_l	1,000 kg/ m^3	a_4	2.13	c_2	1.49
c_o	7 mg/kg	a_5	0.961	c_3	-1.76×10^{-2}

Once the pdfs of the estimated values $\hat{c}_\infty$ are obtained, the bounds of the uncertain parameter c_∞ with different possible values of d_m can be determined by using the results in Figure 6. For instance, when d_m is equal to 100 μm , the estimated pdf by the KDE method is shown in Figure 7. In this figure, $\gamma = 0.5\%$ is used, the lower bound $c_{\infty L}^U$ is 0.24 and the upper bound $c_{\infty U}^U$ is 0.29. As a result, the central point $c_{\infty C}^U$ in Figure 7 can be calculated, which is 0.265 in this case.

After determining the bounds of the uncertain parameter, the scenario tree can be generated. In Figure 8, the blue region is determined by empirical estimation and the box uncertainty region is adopted. Under this situation, we only know the rough bounds of each uncertain parameter through the prior knowledge. The red region is identified by the historical data using the proposed method. This region is much smaller than the blue one, since the correlations between the uncertain parameters are considered. Moreover, the blue region may contain several unlikely-to-happen scenarios. For example, $\mathbf{d} = [60, 0.29]^T$ (the point at the top left) is included in the blue region.

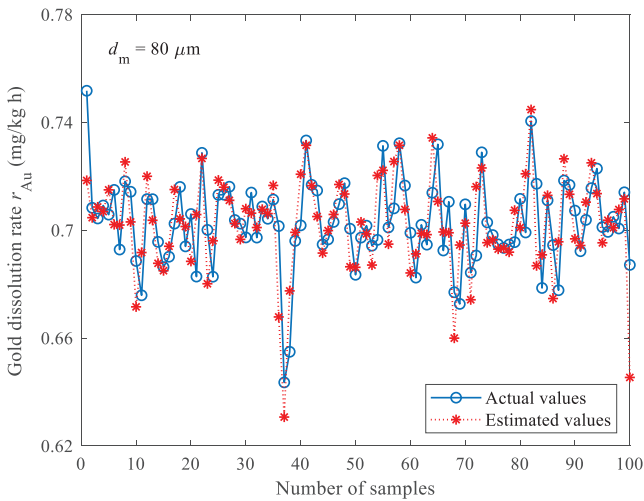


FIGURE 5 Comparison of the actual values and estimated values of the gold dissolution rate [Color figure can be viewed at wileyonlinelibrary.com]

Generally, smaller average diameter of the ore particles is beneficial to extract the gold from the ore. Such a large residual concentration is inconsistent with the actual situation. Thus, including these unlikely-to-happen scenarios may result in conservative solutions, and lead to economic loss.

Finally, by numerically integrating the pdf in Figure 7, the quantile function in Equation (26) can be obtained. In this case, $p(c_{\infty L}^U)$ is 0.0670, $p(c_{\infty C}^U)$ is 0.7953, and $p(c_{\infty U}^U)$ is 0.1377. If $p(d_m^L) = 0.25$, $p(d_m^C) = 0.5$, and $p(d_m^U) = 0.25$ are assumed in this work, the probabilities for each scenario in Figure 8 can be easily computed, which will be further used in the resulting large NLP (34). It should be noted that, the average diameter of the ore particles d_m is generally determined by the operating states of the ore comminution and grinding subprocess and the properties of the ore, and it is difficult to keep d_m unchanged in industrial plant. To consider the uncertainty of d_m and reduce the size of the NLP, three possible values of d_m (the minimum, average, and maximum values) are considered as shown in Figure 8.

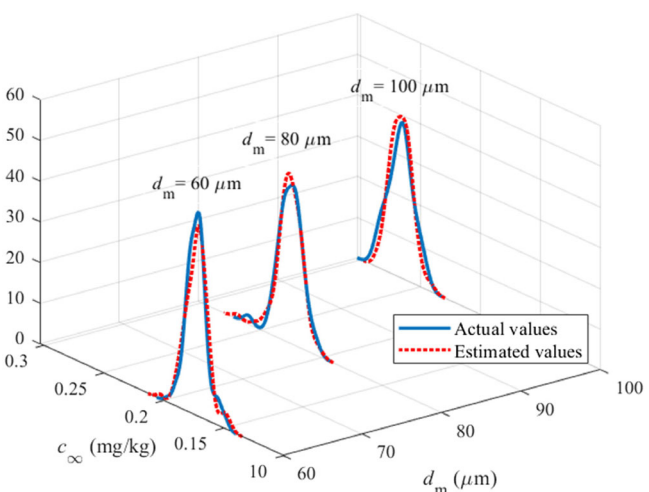


FIGURE 6 Comparison of the probability density functions for the actual and estimated values of the residual gold concentration in the solid phase [Color figure can be viewed at wileyonlinelibrary.com]

	Finite difference	Polynomial fitting	Tikhonov regularization
RMSE	0.0638	0.0429	0.0127
MAE	0.2513	0.1116	0.0417

TABLE 2 Comparison of the estimation results of different derivative estimation methods with the condition $d_m = 80 \mu\text{m}$

TABLE 3 The estimating results for the gold dissolution rate and residual gold concentration in the solid phase with different possible values of the average diameter of the ore particles

	$d_m = 60 \mu\text{m}$		$d_m = 80 \mu\text{m}$		$d_m = 100 \mu\text{m}$	
	r_{Au} (mg/kg h)	c_∞ (mg/kg)	r_{Au} (mg/kg h)	c_∞ (mg/kg)	r_{Au} (mg/kg h)	c_∞ (mg/kg)
RMSE	0.0121	0.0065	0.0127	0.0070	0.0134	0.0073
MAE	0.0699	0.0345	0.0417	0.0238	0.0473	0.0261
λ	10^5	—	10^4	—	10^4	—

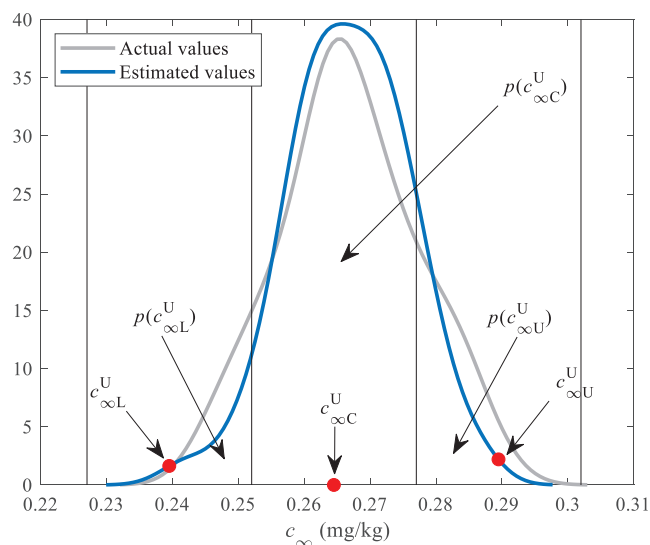


FIGURE 7 The probabilities of the scenarios determined by KDE with the condition $d_m = 100 \mu\text{m}$ [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

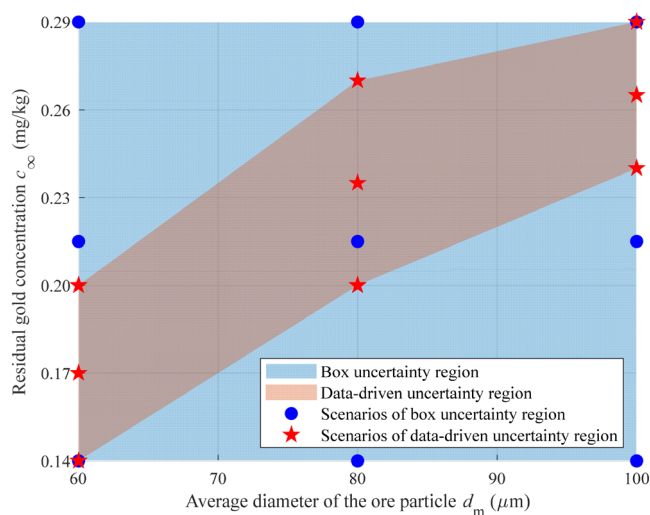


FIGURE 8 Comparison the uncertainty regions and scenarios generated by empirical estimation and historical data. The blue region represents the box uncertainty region determined by empirical estimation, and the red region is the data-driven uncertainty region determined by historical data. The minimum, average and maximum values of each uncertain parameter is used to generate the scenario trees [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Although the residual gold concentration in the solid phase is affected by the average diameter of the ore particles and the properties of the ore, it is impossible to build the empirical model (35) accurately in an industrial GCLP, and in turn to obtain the correlations between the two uncertain parameters. Since the lower and upper bounds of each uncertain parameter can be determined by prior knowledge (empirical estimation), the box uncertainty set (blue region in Figure 8) is a viable option. Even if the uncertain region is enlarged, the box uncertainty set could still generate a relatively reasonable

scenario tree as used in multi-stage MPC.²⁷ One of the merits of the proposed multi-stage EMPC method is that the useful information in the historical dataset is extracted. Although the correlations between the two uncertain parameters remain unknown, a more reasonable uncertainty set (red region in Figure 8) can be formulated to generate the scenario tree and further improve the economic performances.

4.4 | Case 1: Ideal conditions

The economic performances of EMPC for the industrial GCLP in this work can be measured by the total economic cost index,¹⁴ which is defined as follows

$$J_{\text{EMPC}} = \sum_{k=K_0}^{K_f} P_{\text{CN}} q_{\text{CN},1k}^* + P_{\text{dCN}} q_{\text{d},k} c_{\text{CN},4k} + P_{\text{dAu}} q_{\text{s},k} c_{\text{s},4k}. \quad (37)$$

where $c_{\text{CN},4k}$ and $c_{\text{s},4k}$ are the actual states under the control input $q_{\text{CN},1k}^*$ computed by classic EMPC or multi-stage EMPCs with different uncertainty regions, and K_0 and K_f are the initial and final time, respectively. To better evaluate the performances of the proposed multi-stage EMPC, classic EMPC is first implemented as the baseline method in this work. The feeding flow rate and concentration of the slurries are kept unchanged. The dissolved oxygen concentration in the leaching tanks is also constant, and the measurement noise is not appended to the states. The average diameter of the ore particles keeps at its nominal value, that is, $d_m = 80 \mu\text{m}$, and the parameter e_m is equal to zero. Initially, excessive cyanide is added in the first leaching tank to guarantee the leaching rate. The sampling time is 0.1 hr, and the final time is 4 hr. Equation (37) is used as the objective function of classic EMPC, and the leaching rate constraints in Equation (32) are considered as well. The control performance is first shown in Figure 9.

At the first stage, less cyanide solution is added (see Figure 9a), because of the higher cyanide concentration in the leaching tanks. After that, the amount of cyanide input gradually increases, and the production cost also increases at the same time. Finally, the leaching rate reaches the limit, that is, 95% (see Figure 9b). Since the predesigned leaching rate has a higher level, further increasing the leaching rate will result in the loss of economics. To this end, classic EMPC maintain the leaching rate at its predesigned value. The prediction and robust horizons are 10 and 1 in this case, and the total economic cost for classic EMPC is \$1,702.83. Table 4 also summarized the economic performance of the classic EMPC when the prediction horizon is 5. Due to the short prediction horizon, the EMPC method gets the worst objective value. Thus, the multi-stage EMPCs with prediction horizon equaling to 10 are tested in the following case studies.

4.5 | Case 2: Only uncertain residual gold concentration

In order to illustrate the application of the proposed multi-stage EMPC, another case is further considered in this section. The same

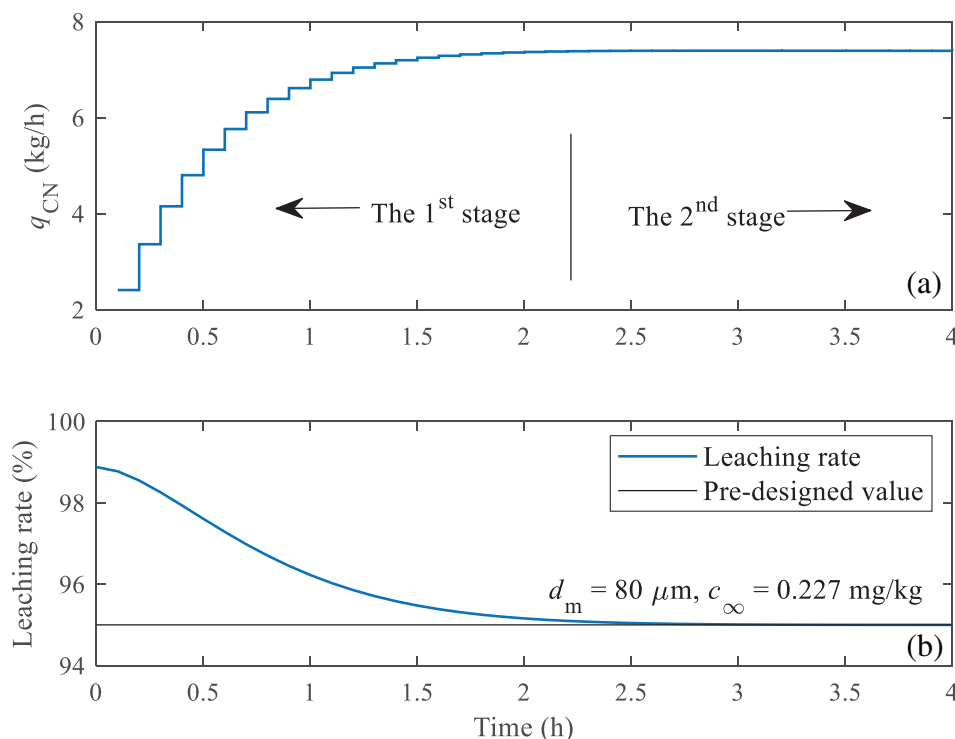


FIGURE 9 The control performances of the classic EMPC for GCLP under the ideal conditions [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 4 Comparison of economic performances of classic EMPC and multi-stage EMPC methods. The PH means prediction horizon, and the Box and Data-driven represent the box uncertainty region and the data-driven uncertainty region in this table

	Classic EMPC		Three-scenario case		Nine-scenario case	
	PH = 5	PH = 10	Box	Data-driven	Box	Data-driven
Case 1	\$4,541.83	\$1,702.83	\$1,768.42	\$1,726.90	\$1,768.34	\$1,765.77
Case 2	Infeasible	Infeasible	\$1,741.83	\$1,725.21	\$1,782.67	\$1,745.21
Case 3	Infeasible	Infeasible	Infeasible	Infeasible	\$2,775.77	\$2,708.17
Comp. Time	0.070 s/iter.	0.137 s/iter.	1.50 s/iter.	1.41 s/iter.	21.83 s/iter.	22.49 s/iter.

conditions as in the previous case are considered. The only difference is that the parameter e_m is not equal to zero, but has a distribution as $N(0, 0.02)$, as in the last case. Three EMPC methods are considered here. The first one is the classic EMPC used in the last case. The other two methods are the proposed multi-stage EMPC with three scenarios (three-scenario case in Table 4). The scenario tree of the former one is obtained by empirical estimation, and the probabilities of each scenario are the same. The latter generates the scenario tree by using the proposed method as shown in the last part of this work. The lowest economic cost (\$1,701.69) is obtained by classic EMPC, because the least cyanide solution is added in the first leaching tank as shown in Figure 10a. However, the leaching rate of the classic EMPC passes the predesigned value during the period from 3 to 3.5 hr (see the black circle in Figure 10b). Since the processed ore with higher gold concentration cannot be treated by the following sub-processes, such operation may lead to the loss of gold, which is usually forbidden in the gold smelting plant.

For the other two multi-stage EMPC methods, both of their leaching rates can meet the requirements of the production. In Figure 11, the curve of the residual gold concentration in the solid

phase is given. From Figure 8, the lower and upper bounds for the multi-stage EMPC with box uncertainty region are 0.14 and 0.29, respectively. However, for the proposed multi-stage EMPC, these bounds are 0.20 and 0.27. The range determined by the historical data is much narrower than the box uncertainty region, and the probabilities for each scenario can be well resolved. In Figure 10a, the control profiles (the flow rates of the cyanide solution) are also compared. Since the proposed multi-stage EMPC with the scenario tree generated by historical data adds less cyanide in the first leaching tank, the economic cost is lower than the other multi-stage EMPC with box uncertainty region. Table 4 also summarized the total economic costs of different methods. The proposed multi-stage EMPC method saves about 1% cost in this short period. And from Figure 10a, the same conclusion can also be achieved. The multi-stage EMPC methods evaluated in this case are also applied to the ideal conditions in the last case. Since three scenarios are considered, the economic costs are higher than that of the classic EMPC. However, if the classic EMPC is used to operate the GCLP in this case study, the feasible solution cannot be obtained (also see Table 4).

FIGURE 10 Comparison of the control performances of classic EMPC and multi-stage EMPC [Color figure can be viewed at wileyonlinelibrary.com]

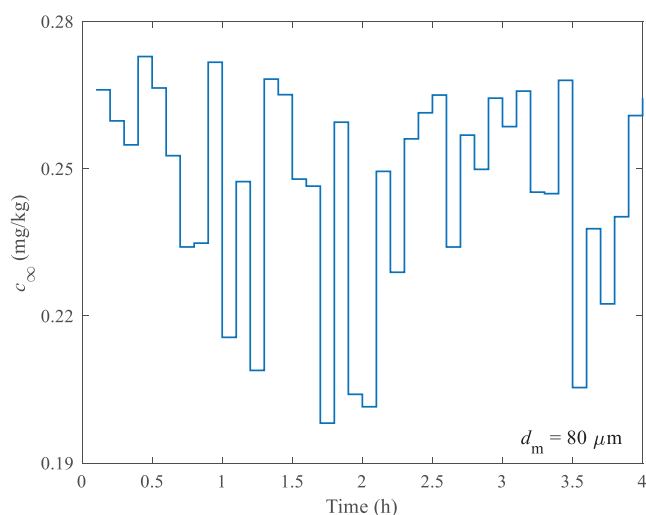
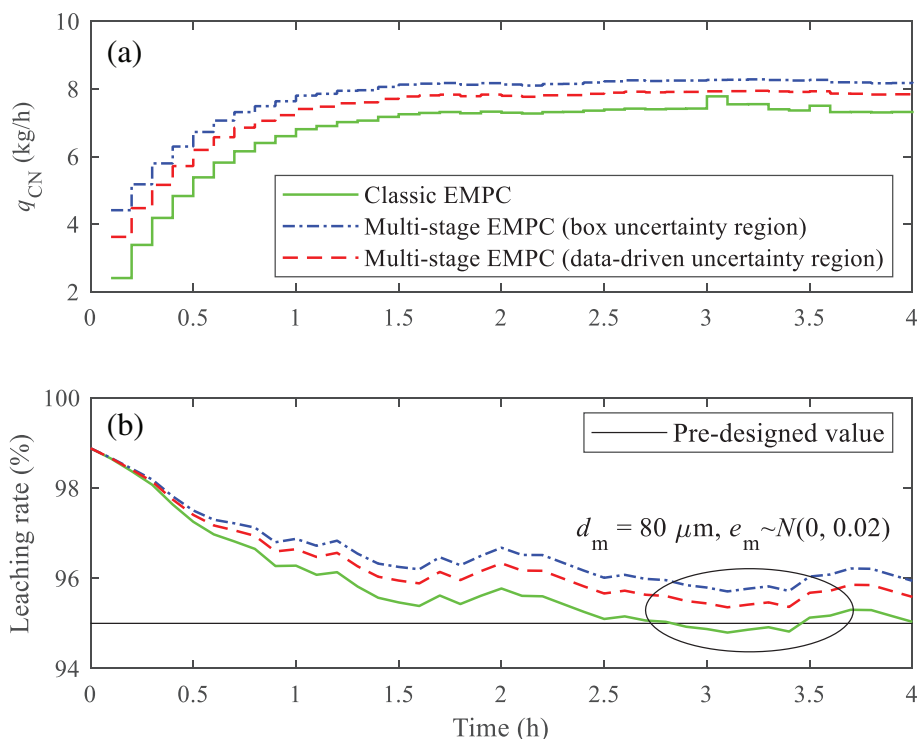


FIGURE 11 The curve of residual gold concentration in solid phase for Case 2 [Color figure can be viewed at wileyonlinelibrary.com]

To further evaluate the influence of the confidence level γ and robust horizon RH, the economic performances with different values of γ and RH are also illustrated in Table 5. Because the extreme values are not included in the scenario tree with $\gamma = 5\%$, the proposed method cannot obtain robust constraint satisfaction. When $\gamma = 0\%$ is used, the control performances are slighter worse than that of $\gamma = 0.5\%$, but the economic performances are better than the multi-stage EMPC with box uncertainty region. A possible reason is that the correlations between the uncertain parameters are considered as well. For an industrial GCLP, $\gamma = 0.5\%$ is used in this work to reduce the

TABLE 5 Comparison of economic performances of the proposed multi-stage EMPC methods with different values of confidence level γ and robust horizon RH

	$\gamma = 5\%$	$\gamma = 0.5\%$	$\gamma = 0\%$	Comp. Time
RH = 1	Infeasible	\$1,725.21	\$1,731.39	1.41 s/iter.
RH = 2	Infeasible	\$1,721.06	\$1,727.23	25.11 s/iter

effects of the outliers and include the extreme values as much as possible. When RH increases from 1 to 2, the economic performances only slightly improved. Nevertheless, the number of decision variables and the computation time significantly increase from 402 and 1.5 s to 1,272 and 25.11 s, respectively. In order to achieve a balance between the computational burden and economic performances, RH = 1 is considered in the following example.

4.6 | Case 3: A simulated industrial GCLP conditions

In this section, a simulated industrial case is studied. In this case, the initial conditions are the same as the previous two cases. However, the uncertain parameter d_m has three possible values, and the measurement noise is appended into the states as in the last case. In addition, the feeding flow rate of the solid and liquid phases are also changed frequently to simulate the disturbances that come from the upstream floatation process. Three multi-stage EMPC approaches are considered to dynamically operate the GCLP. The first one, as used in Case 2, has a scenario tree with three branches generated by historical data, that is, the minimum, average, and maximum values of the

uncertain residual gold concentration in the solid phase determined by historical data (the red star in the green cycle in Figure 12b). The second multi-stage EMPC has nine scenarios, and the box uncertainty region is denoted by blue as shown in Figure 12b. In that figure, the blue points represent the scenarios, which is also generated by considering the minimum, average, and maximum values of the uncertain average diameter of the ore particles and residual gold concentration in the solid phase in the blue region. The last multi-stage EMPC method has nine scenarios as well, and the data-driven uncertainty region is denoted by red as shown in Figure 12b. Each red star in that figure represents one scenario, which is obtained by the proposed method in the last section.

In GCLP, the average diameter of the ore particles may vary due to the fluctuation of the upstream process. When d_m increases, it is difficult to extract gold from the ore, and the satisfaction of the leaching constraints cannot be guaranteed by using the classic EMPC (see Table 4). Thus, in this case, d_m is assumed to change randomly every 2 hr, and c_∞ will change accordingly. The trend of c_∞ is drawn in Figure 13. The economic costs of the different methods are compared in Table 4. Although the multi-stage EMPC with three scenarios adds the lowest amount cyanide most of the time as shown in Figure 12a, the corresponding leaching rate drastically passes the predesigned value (from 2 to 4.5 hr). This is because only $d_m = 80 \mu\text{m}$ is considered. When the uncertain parameter d_m returns to $80 \mu\text{m}$, due to the insufficient cyanide added in the early stage, the leaching rate may slightly violate the constraint (see Figure 12d).

The multi-stage EMPC methods with nine scenarios (nine-scenario case in Table 4) are also considered. The multi-stage EMPC

with box uncertainty region is firstly utilized, and the total economic cost is \$2,775.77. The leaching rate is much higher than the predesigned value. As shown in Figure 12a, the most amount of cyanide is added. When the proposed multi-stage EMPC with a data-driven uncertainty region is applied, the best dynamic operating performance is obtained, and the total economic cost is \$2,708.17. Nearly 2.5% cost is saved in this period. This is because that the

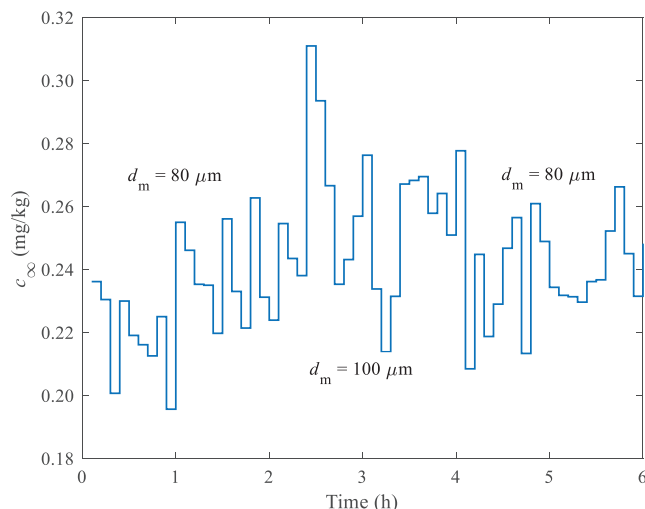


FIGURE 13 The curve of residual gold concentration in solid phase for Case 3 [Color figure can be viewed at wileyonlinelibrary.com]

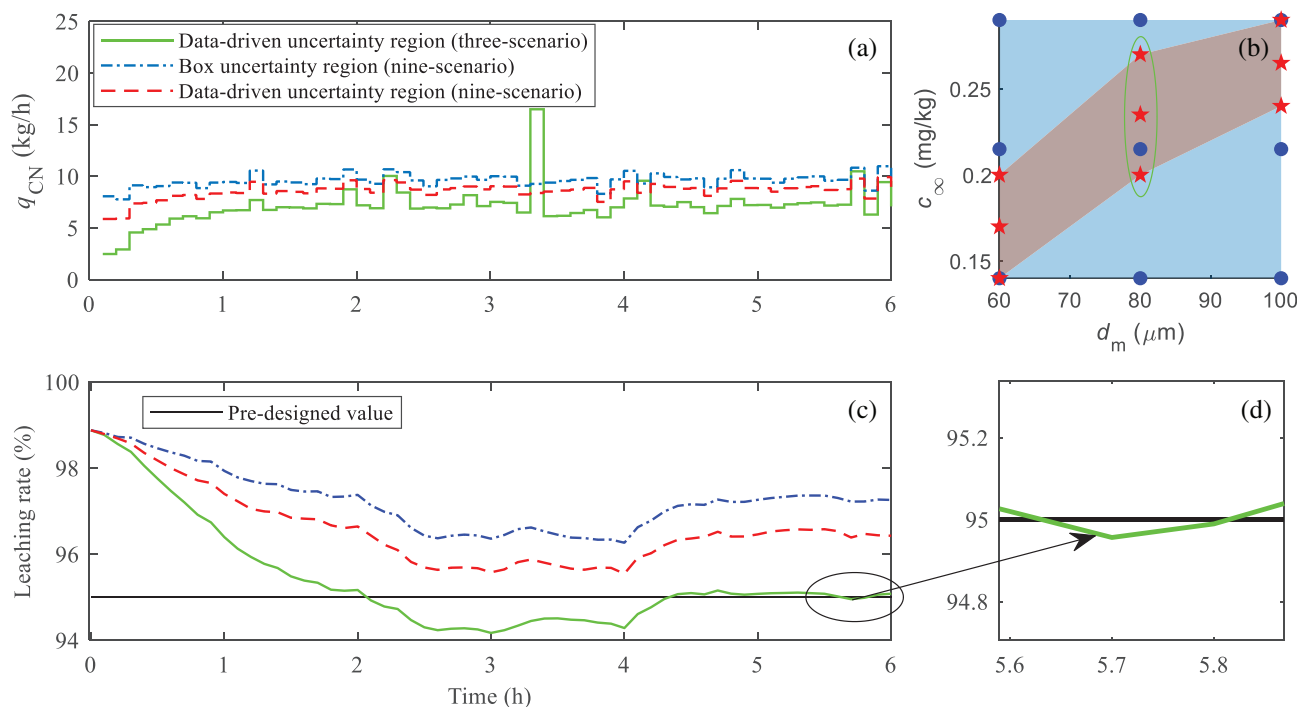


FIGURE 12 Comparison of the control performances of different multi-stage EMPC methods, including the scenario tree with three branches generated by historical data, the scenario tree with nine branches generated by empirical estimation, and the scenario tree with nine branches generated by historical data [Color figure can be viewed at wileyonlinelibrary.com]

proposed method properly designs the uncertainty region and determines the probability of each scenario.

Finally, the computation time for each method is considered in Table 4 as well. For classic EMPC, the formulated NLP has only 77 (for prediction horizon is 5) or 142 (for prediction horizon is 10) decision variables, and the number of constraints is 65 and 70, respectively. Thus, the computation time of each iteration is very short, and generally less than 1 s. However, when the multi-stage EMPC methods are considered, the number of decision variables in the formulated NLP increases drastically. For multi-stage EMPC with three scenarios, there are 402 continuous variables and 392 constraints in this NLP problem, and the average computation time of each iteration is around 1.5 s. For multi-stage EMPC with nine scenarios, the NLP has 1,182 decision variables and 1,178 constraints, and the average computation time increases to about 22 s.

5 | CONCLUSIONS

In order to achieve both transient and steady-state economic optimization of GCLP simultaneously, a multi-stage EMPC for GCLP was presented in this work. The proposed strategy included two steps. The first one was to estimate the unmeasurable uncertain parameters by using the historical data; the second one was to generate the scenario tree and formulate the mathematical optimization problem for GCLP at each sampling point. To restrain the noise in the measurements from being amplified and propagated into the parameter estimation, a Tikhonov regularization based method was used to estimate the reaction rates in GCLP in the first step, and then this uncertain parameter can be obtained by solving the nonlinear equations integrated with prior knowledge. To further improve the economic performances of GCLP, the estimating results integrated with the KDE method were adopted to design the uncertainty region and generate the scenario tree for multi-stage EMPC in the second step. The probability of each scenario could also be well settled. The NLP for the proposed multi-stage EMPC were formulated to directly optimize the cost function of GCLP, while accounting for the leaching rate constraints. The proposed method was applied to a simulated industrial GCLP.

The merits of the proposed multi-stage EMPC can be summarized as follows: Compared with classic EMPC, the proposed multi-stage EMPC can guarantee the satisfaction of leaching rate constraints under uncertainty. Lower leaching rates may result in the loss of gold, which is usually unacceptable in the gold smelting plant. By comparing the multi-stage EMPC with the box uncertainty region, the proposed method obtains less conservative solution and consumes less cyanide. Since the cyanide is toxic, excessive cyanide would have an adverse effect on the environment. Over the long operating horizon, the accumulated benefits could be substantial.

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NOTATION

$c_{CN,l}$	The cyanide concentration in the liquid phase of the l th leaching tank
$c_{s,l}, c_{l,l}$	The gold concentration in the solid and liquid phases of the l th leaching tank
$c_{O,l}$	The dissolved oxygen concentration in liquid phase of the l th leaching tank
$c_{w,l}$	The mass concentration of the slurry in the l th leaching tank
c_{∞}	The residual gold concentration in the solid phase
$\mathbf{c}$	The states of GCLP
d_m	The average diameter of the ore particles
$\mathbf{d}$	The realization of the uncertainty
$m_{s,l}, m_{l,l}$	The amounts of the solid and liquid phases in the l th leaching tank
$q_{CN,l}$	The flow rate of the cyanide solution added into the l th leaching tank
$q_{s,l}, q_{l,l}$	The feeding flow rate of solid and liquid phases of the l th leaching tank
$r_{Au,l}$	The gold dissolution rate of the l th leaching tank
$r_{CN,l}$	The cyanide consumption rate of the l th leaching tank
$t_{r,l}$	The average residence time of the l th leaching tank
$\mathbf{u}$	The control inputs of the discretized process model
v_l	The net reactor volume of the leaching tank of the l th leaching tank
$\mathbf{x}$	The states of the discretized process model
$\mathbf{x}_{sp}$	The desired set-point of the states
$y_{s,l}$	The generalized gold concentration in solid phase of the l th leaching tank
M	The number of available samples in the historical dataset
N_s	The number of scenarios
N_p	The prediction horizon
S	The set of the scenario
P_{CN}	The price of the cyanide solution
P_{dCN}, P_{dAu}	The price for disposing the residual cyanide and gold
$\mathbf{Q}, \mathbf{R}$	The weight matrices corresponding to the states and control inputs
$\mathbb{X}, \mathbb{U}$	The sets of the states and control inputs of the discretized process model

GREEK LETTERS

ρ_s, ρ_l	The densities of the solid and liquid phases
ω	The probability of the scenario
ζ	The set of residual gold concentration in solid phase
ξ	The set of scenario tree

SUBSCRIPTS

k	The sampling time
L	The minimum value

- C The average value
U The maximum value

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